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ORIGINAL RESEARCH

Power system instability prediction from the solution pattern of differential Riccati equations

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Abstract

Power system stability characteristics are typically evaluated in terms of small- and large-signal (transient) stability. Access to the time-varying A-matrix of a state-space-based power systems model during transient conditions can be utilized to apply linear time-varying system concepts for large-signal stability analysis. In linear time-varying system analysis, the differential Riccati equation (DRE) plays a vital role when the power system is subjected to a severe disturbance. The Möbius transformation is proposed in this paper to solve the DRE with singularity issues. It is shown that the solution of the DREs follows a specific mathematical pattern when the power system is stable but does not follow this pattern when the system progresses toward instability. The proposed method can be used in large-signal stability analysis to predict instability and make the stability analysis more efficient. Additionally, the vector-DRE is proposed to generalize the index in a large-scale power system. Results show that analyzing the corresponding Riccati equation's behaviour can help researchers predict a power system's performance and improve the control and management of the system.

1 | INTRODUCTION

Small-signal and large-signal analysis are two common approaches to examine different aspects of power system stability. For small-signal stability analysis the nonlinear system equations are linearized around an operating point, and then linear methods are used to complete the analysis [1–3]. The analysis can be based on various methods such as: transfer functions (closed/open loop, single/multi-input, and single/multi-output) [4], eigenvalue extraction from the state-space model of the system [5], and impedance-based analysis by Nyquist theory [6]. However, the power system is not always linear, so it should be evaluated under nonlinear conditions when a severe (large-signal) disturbance occurs. Lyapunov-based methods are direct methods that examine a power system's stability without solving the differential equations [7–9]. However, obtaining an appropriate Lyapunov function is complicated in a large-scale power system with complex components. Time-domain simulation by a numerical integration method is currently the most widely used approach for transient stability analysis. Differential and algebraic equations of the power system are solved by a numerical method, and transient indices are derived from the time-domain response [10–12].

The A-matrix that can be extracted from a state-space model during the transient condition of a power system is time-varying. Thus, the conventional concept of the eigenvalue does no longer determine the stability of the system. Furthermore, the traditional method for detecting instability by monitoring the excursion of rotor angles beyond $\delta = 180°$ does not provide an efficient index for large-signal stability analysis based on numerical integration methods. Therefore, the possibility for obtaining an early prediction of instability (before the power angle exceeds the limit of instability) motivates us to search for a novel method based on time-varying A-matrix properties. According to LTV system theory, the differential Riccati equation (DRE) is a characteristic equation of the system when arrays of A-matrix of the system change throughout the time. DRE is studied theoretically and numerically in this paper to predict the dynamic performance of the power system.

DRE is used in different areas of mathematics and physics and also appears in many applied problems. In recent years, the application of Riccati-type equations has expanded beyond

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traditional analyses, playing a pivotal role in the stability assessments of complex systems, such as switched linear systems. It is noteworthy that switched systems can be considered approximations of linear time-varying (LTV) systems, where the time-varying elements of the A-matrix change in discrete steps rather than continuously [13]. In [14] an adaptive control design utilizes DRE to manage parametric uncertainties and disturbances effectively. This approach involves the derivation of a Lyapunov function from the closed-form solutions of a family of DREs, tailored to ensure stability across finite switching intervals.

In this paper, the solution of DREs is examined in different conditions to give an instability prediction index for large-signal stability analysis. According to the mathematical equations and the monitored excursion of the DRE's solution, it is shown that the solution corresponding to a power system follows a specific pattern of a tangent under steady-state and in stable conditions. However, out-of-pattern solution emerge when the system progress toward unstable condition. These differences are the foundation of the method proposed in this paper for predicting power system transient instability. The Möbius method is used to solve DRE numerically with singularities. Singularities limit the power system stability analysis from accessing different solutions of DRE by creating a numerical instability. The Möbius transform tackles singularities by changing the coordinate system and doing three mapping operations (rotation and dilation, translation, inversion). Moreover, this paper studies power systems with state-space model dimensions greater than two by generalizing the Riccati equation to the vector-differential Riccati equation (V-DRE). The A-matrix corresponding to the power system is gradually broken down into smaller matrices by solving V-DRE, and then the proposed instability prediction method is applied to the smaller matrices individually.

Numerical methods could be used to calculate A-matrix of the power system. The methods have different advantages and disadvantages concerning aspects of stability, accuracy, and computational burden [15]. Numerical methods are also generally categorized into explicit and implicit methods. In explicit methods, the integration formula is applied directly, and states are calculated from only the last time step. However, in implicit methods, the current time step is also involved, and the equation is solved iteratively. Implicit methods with strong numerical stability are more suitable for numerical simulation of a complex system. Gear's method is used in this paper as a numerical integration method. The method is composed of two stages: the predictor and corrector stages. The states are predicted explicitly in the predictor stage, and then the predicted values are modified implicitly in the corrector stage [16, 17]. In the correction step of Gear's method, the partial differentiation of algebraic and differential equations is used to form the power system's state matrix. Thus, the method inherently provides access to the time-varying A-matrix during time-domain simulations and offers a new approach for predicting power system instability. On this basis, the contributions of this paper can be summarized as:

- Proposing a new application of DRE in stability analysis of power systems.
- Prediction of instability of the power system in large-signal stability analysis.
- Development of an approach for effectively handling the singularities of DRE with complex values.
- Formulation of a method for efficiently managing instability detection in power systems with state-space models of order greater than two.

The aforementioned contributions will be utilized in the field of stability analysis to enable earlier predictions. These contributions will be demonstrated through three different cases in the simulation results.

2 | GEAR'S METHOD

Consider a general nonlinear DAE as

$$y' = f(y, x, t), \quad 0 = g(y, x, t) \tag{1}$$

where y is the state, f is a differential function, x is an algebraic variable and g is an algebraic function. The prediction stage of Gear's method is given by the following equations:

$$y_{n+1}^{P} = y_n + H_{n+1} y_n' + H_{n+1}^2 y_n''/2 \tag{2}$$

$$y_{n+1}'^{P} = y_n' + H_{n+1} y_n'' \tag{3}$$

$$y_{n+1}''^{P} = y_n'' \tag{4}$$

where H is the integral step size, and y_{n+1} is the next step solution [18]. In the correction stage of the method, the predicted values are corrected as

$$y_{n+1} = y_{n+1}^{P} + \Delta y \tag{5}$$

$$y_{n+1}' = y_{n+1}'^{P} + \Delta y\, I_1/H_{n+1} \tag{6}$$

$$y_{n+1}'' = y_{n+1}''^{P} + 2\Delta y\, I_2/H_{n+1}^2 \tag{7}$$

where I_1 and I_2 are constant values, which depend on the integral step size. Δy is the difference between the predicted and the corrected values. To solve the differential equation presented in Equation (1), the method defines a new function U based on Equations (1) and (3) and the system of DAE is converted into a system of ordinary differential equation (ODE) by differentiating it as

$$\begin{bmatrix} U_{n+1} \\ g_{n+1} \end{bmatrix} = \begin{bmatrix} I_1 - H\dfrac{\partial f}{\partial y} & -H\dfrac{\partial f}{\partial x} \\ \dfrac{\partial g}{\partial y} & \dfrac{\partial g}{\partial x} \end{bmatrix} \begin{bmatrix} \Delta y \\ \Delta x \end{bmatrix} \tag{8}$$

Since U_{n+1} and g_{n+1} will become zero, the Newton–Raphson method is used to calculate the changes of variables (Δx and

Δy). According to Equation (8), the Gear's method will calculate the partial derivatives of the system model in every integration step. Therefore, the A-matrix is obtained as [17]:

$$A = \frac{\partial f}{\partial y} - \left(\frac{\partial f}{\partial x} \left(\frac{\partial g}{\partial x} \right)^{-1} \frac{\partial g}{\partial y} \right) \tag{9}$$

Access to the A-matrix of a power system during both steady-state and dynamic conditions opens new doors in the stability analysis of a power system during transient conditions. It is worth noting that both continuous-time and discrete-time equations are used in this paper driven by the necessity to apply theoretical concepts effectively. This dual approach enables us to harness the analytical strengths of continuous-time modelling for theoretical development while capitalizing on the advantages of discrete-time methods for numerical integration.

3 | ANALYSIS OF A POWER SYSTEM UNDER TIME-VARYING CONDITIONS

The A-matrix extracted by Gear's method changes over time, for which the LTV theory can be applicable. Therefore, at each step we have:

$$Y'(t) = A(t)\, Y(t) \tag{10}$$

$$A(t)\, e(t) = \lambda(t)\, e(t) + e'(t) \tag{11}$$

where $Y = [y_1, y_2, \ldots y_n]$ is the vector of the states, $\lambda(t)$ is the dynamic eigenvalue and $e(t)$ is the dynamic right eigenvector. The calculated dynamic right eigenvector forms a dynamic similarity matrix ($S(t) = [e_1(t), e_2(t), \ldots, e_n(t)]$) by which the A(t) is transformed to a diagonal matrix as:

$$\Lambda = S^{-1}(t) A(t) S(t) - S^{-1}(t) S'(t) \tag{12}$$

The dynamic eigenvalues will be on the diagonal of Λ. To calculate the $S(t)$ in an LTV system a function (Ricatti) is required that satisfies Equation (11) [19]. In a second-order LTV system with state matrix ($A = [a_{11}(t), a_{12}(t); a_{21}(t), a_{22}(t)]$), the following differential Ricatti equation should be solved.

$$L'(t) = -a_{12}(t) L^2(t) - (a_{11}(t) - a_{22}(t)) L(t) + a_{21}(t) \tag{13}$$

where $L(t)$ is the Ricatti function. The differential Riccati equation presented in Equation (13) is solved by the numerical integration method. However, in a power system with complex eigenvalues, the corresponding Riccati equation has singularities which the standard methods or Gear's method cannot deal with. One of the singularities is in the root of the algebraic Riccati equation ($L'(t) = 0$ in Equation (13)) as:

$$L_0 = \frac{-(a_{11} - a_{22}) + \sqrt{(a_{11} - a_{22})^2 + (4 a_{12} a_{21})}}{2 a_{12}} \tag{14}$$

Stability analysis of a power system will fail if a numerical integration procedure passes through the singularities. It means that an unstable excursion is predicted for a stable power system. To show the problem consider an A-matrix of an LTV system as:

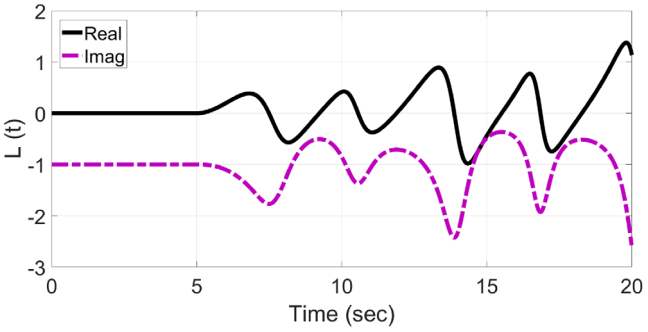


FIGURE 1 Solution of Riccati equation with complex initial value by ODE15s solver.

$$A = \begin{bmatrix} 0 & a_{12} \\ 1 & 0 \end{bmatrix} \tag{15}$$

$$a_{12} = -1 \quad t < 5$$

$$a_{12} = -1 + 0.5 e^{-0.01(t-5)} \sin(t-5) \quad t \geq 5$$

which is a time-invariant system for $0 \leq t < 5$, and time-varying after $t = 5$ s. The corresponding DRE of the system (i.e. Equation (13)) is solved using the ODE15s (it is a function used to solve stiff differential–algebraic equations in MATLAB), and the result is shown in Figure 1. The results show that the Riccati equation's solution has a constant complex value $L = -1i$ during $0 \leq t < 5$. However, for $t \geq 5$, the solution has unstable oscillations even when the time-varying array (a_{12}) returns to its initial values (-1) since the exponential function has a negative time constant. The presented example shows that the Riccati equation has a singularity in the root of the algebraic Riccati equation ($L = -1i$), which will make the power system stability analysis to fail. Therefore, it is required to use an appropriate numerical method to analyse the power system's stability by DRE.

4 | INSTABILITY PREDICTION BY EVALUATING A SOLUTION OF DRE

The previous section shows that standard methods will be unable to integrate through the Riccati equation's singularities. However, one of the properties of the Riccati equation is the existence of movable singularities, which is an artefact of the coordinate system. Therefore, one can integrate through singularities by changing the coordinate system. The Möbius method is a transformation that can change the coordinate system and has the following form [20]:

$$f(z) = \frac{\alpha z + \beta}{\lambda z + \gamma} \tag{16}$$

where z is a complex value, and $\alpha, \beta, \lambda, \gamma$ are complex coefficients. According to (16), the Möbius transformation consists of three types of mapping, defined as rotation and dilation ($z \to \alpha z$), translation ($z \to z + \beta$), and inversion ($z \to \frac{1}{z}$) which will change the coordinate system of an equation. Therefore, the Möbius method can deal with numerical instability and pass accurately through the singularities by changing the coordinate system. The Möbius method's coefficients are obtained from the coefficients of the DRE presented in Equation (13) as:

$$\alpha = 1 + (H \times a_{11}), \quad \beta = H \times a_{12} \qquad (17)$$
$$\lambda = H \times a_{21}, \quad \gamma = 1 + (H \times a_{22})$$

where, H is the step size of the integration. Therefore, the solution of the next step of DRE is calculated as:

$$L_{i+1} = \frac{\alpha L_i + \beta}{\lambda z + \gamma} \qquad (18)$$

where i is the counter of integration steps. DRE can be effectively integrated via an explicit, one-step numerical method. Passing through the Riccati equation's singularities by the Möbius method provides an opportunity to focus on other possible solutions of a DRE with different initial values. Generally, finding a theoretical solution of a DRE is complicated, and that is why numerical integration methods are used in the literature. However, it is possible to demonstrate that Equation (19) is a solution of DRE presented in Equation (20).

$$L(t) = \sqrt{\frac{a(t)}{b(t)}} \tan\left(\sqrt{a(t)b(t)}t\right) \qquad (19)$$

$$L'(t) = a(t) + (b(t) \times L^2(t)) \qquad (20)$$

According to Equation (19), the DRE presented in Equation (20) has a specific solution pattern of a tangent [21]. This conclusion will be used in this paper for the prediction of the instability of a power system. As an index, a second-order system with a corresponding differential Riccati equation presented in Equation (20) is stable if the DRE solution has a tan pattern, while it is unstable without a pattern of tan. Accordingly, the proposed index can be used in large-signal stability analysis of a power system. To generalize the proposed index into a large-scale power system, the vector differential Riccati equation is used, which will be presented in the next section.

5 | VECTOR DIFFERENTIAL RICCATI EQUATION

This section shows how the proposed instability prediction method can be used for a large-scale power system. There are many components in a large-scale power system, and the extracted A-matrix of the system by the Gear's method has a large dimension. Vector differential Riccati equation (V-DRE) is proposed in this paper to make a proposed method applicable with large-dimension A-matrix [22]. A large dimension can be gradually decreased in different iterations to decompose a large A-matrix into individual smaller matrices or scalars corresponding to different eigenvalues (complex or real value). Consider an A-matrix with dimension k corresponding to a large-scale power system that is extracted by the Gear's method as:

$$y'_k(t) = A_k(t) y_k(t) \qquad (21)$$

In any iteration, the A-matrix ($A_k(t)$) is partitioned as:

$$A_k = \begin{bmatrix} D_k & b_k \\ c_k^T & d_k \end{bmatrix} \qquad (22)$$

According to the partitioned A-matrix presented in (22), the V-DRE is formed as:

$$L'_k = \left(-L_k^T D_k\right) + \left(c_k^T - L_k b_k L_k^T\right) + \left(d_k L_k^T\right) \qquad (23)$$

where L_k is the solution of V-DRE. To decompose the A-matrix, a similarity transformation and its derivative are formed as:

$$P_k = \begin{bmatrix} I & 0 \\ L_k & 1 \end{bmatrix} \qquad (24)$$

$$P'_k = \begin{bmatrix} 0 & 0 \\ L'_k & 1 \end{bmatrix} \qquad (25)$$

According to the similarity transformation and Equation (23), the A-matrix (A_k) is triangularized to Λ_k as:

$$\Lambda_k = P_k^{-1} A_k P_k - P_k^{-1} P'_k = \begin{bmatrix} A_{k-1} & b_{k-1} \\ 0^T & \lambda_k \end{bmatrix} \qquad (26)$$

According to the Equation (25), the K_{th} eigenvalue ($\lambda_k(t)$) of a linear time-varying (LTV) system is on the diagonal of Λ_k, which can be decomposed individually from the rest of eigenvalues. Additionally, a new and smaller A-matrix (A_{k-1}) is on the diagonal, which contains the remaining eigenvalues. λ_k and A_{k-1} can also be calculated as:

$$\lambda_k = d_k - L_k^T b_k \qquad (27)$$

$$A_{k-1} = D_k + b_k L_k^T \qquad (28)$$

Therefore, the dynamic performance corresponding to Kth eigenvalue can be evaluated individually and decoupled from other eigenvalues. By extracting one dynamic eigenvalue, the dimension of the system is reduced by one. The extracted A_{k-1} from the previous iteration is a new input (A-matrix) to the next iteration, and the same procedure (Equations (22) to (27)) is repeated until the final A-matrix becomes a second-order system or a scalar. In this way, a large scale A-matrix is decomposed into the smaller time-varying matrices and so the proposed method for instability prediction is applicable for them.

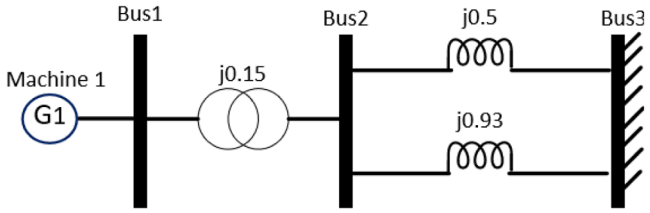


FIGURE 2 Power system-SMIB.

6 | SIMULATION RESULTS

Different case studies are examined in this section to show the application of the DRE, V-DRE, Möbius transformation, and the proposed instability prediction method. Firstly, a single machine to an infinite bus (SMIB) is studied to show the performance of the Möbius method and the proposed index in a second-order system. In the second section, SMIB is equipped with an automatic voltage controller (AVR) and power system stabilizer (PSS) to form a sixth-order system. The second section analyses the system to evaluate V-DRE and the proposed index with a larger A-matrix. Finally, HVDC is integrated into the SMIB in the third section, and the proposed method is evaluated in a more complex power system.

6.1 | SMIB (second-order system)

In this section, the performance of the proposed method is evaluated when a single machine is connected to an infinite bus ($V_{\text{Bus3}} = 0.90081\angle 0$) in a network shown in Figure 2 [5]. The generator (G1) shown in Figure 2 produces an apparent power $S_{\text{Bus1}} = 0.9 + j0.436$ (in per-unit) at Bus1 with voltage $V_{\text{Bus1}} = 1.0\angle 28.34°$ in the steady state. The generator is represented by a second-order model with an internal voltage ($E'\angle\delta$) behind a transient reactance $X'd = 0.3$ p.u., inertia constant $H_{\text{int}} = 3.5$ s, and damping factor $K_{\text{D}} = 0.25$. The differential equations of a second-order system (swing equations) are:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Delta\omega_{\text{r}} = \frac{0.5}{H_{\text{int}}}(P_{\text{m}} - P_{\text{e}} - K_{\text{D}}\Delta\omega_r)$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\delta = \omega_0\,\Delta\omega_r \tag{29}$$

where ω_{r} is the angular frequency of rotor, $P_{\text{m}} = 0.9$ is the mechanical power, P_{e} is the electrical power, w_0 is the base angular frequency, and δ is the power angle of the generator. The network is modelled by the current injection method [23], and a three-phase fault is modelled by a zero shunt impedance to stress the power system under different conditions. The Gear's method is applied to the system, and the system is evaluated in two cases: a stable and an unstable situation.

6.1.1 | Stable condition

A three-phase fault is created at $t = 1$ s close to the Bus2 (Figure 2), and the fault is cleared after 0.05 s by removing the transmission line with impedance $0.93j$. Accordingly, the system is simulated, and the power angle of generator δ is shown in Figure 3a. The simulation shows that the power system is stable if the fault is removed after 0.05 s. The corresponding dynamic characteristic equation (differential Riccati equation) is also evaluated by the proposed method. Gear's method extracts the A-matrix of the system in each step of integration to form DRE of the power system. The extracted A-matrix is:

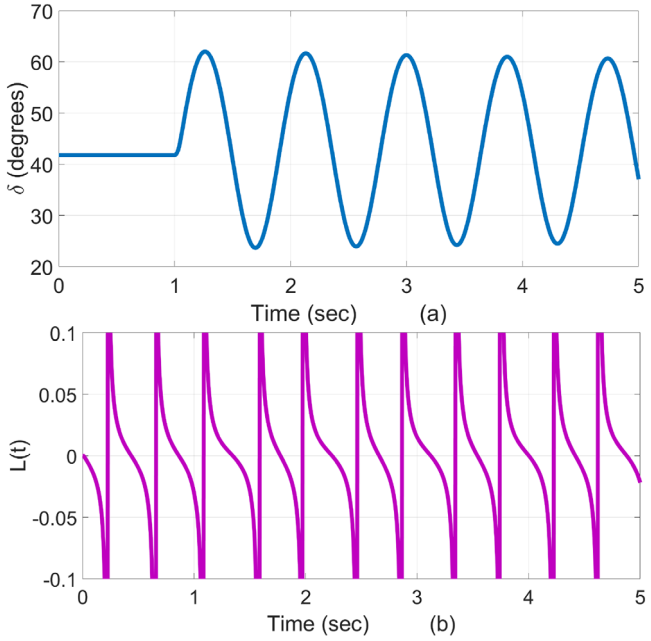


FIGURE 3 Rotor angle responses (a), solution of Riccati equation in stable condition in SMIB (b).

$$A = \begin{bmatrix} a_{11} = -0.0357 & a_{12}(t) \\ a_{21} = 376.99 & a_{22} = 0 \end{bmatrix} \tag{30}$$

The matrix presented in Equation (30) shows that three arrays are constant (since they are just related to parameters of the generator), but the array a_{12} is time-varying and changes based on the power system conditions. Accordingly, the second-order DRE is formed based on Equation (13) as:

$$L'(t) = -a_{12}(t)L^2(t) + 0.0357L(t) + 376.99 \tag{31}$$

The corresponding DRE has singularities in the steady-state solution ($L'(t) = 0 \rightarrow L_0 = -0.1241 - 51.1756i$) and in the zero value ($L_0 = 0$), which cannot be integrated by the standard ODE solvers as explained in Section 3 and shown in Figure 1. Therefore, the Möbius method is applied to Equation (30) to integrate through the singularities. The solution of DRE when using the Möbius method is shown in Figure 3b. As expected (presented in Section 4), the DRE's solution follows a tangent pattern before and after the fault ($t = 1$), which shows that the system is stable. Therefore, the solution of DRE reflects the stable behaviour of the system.

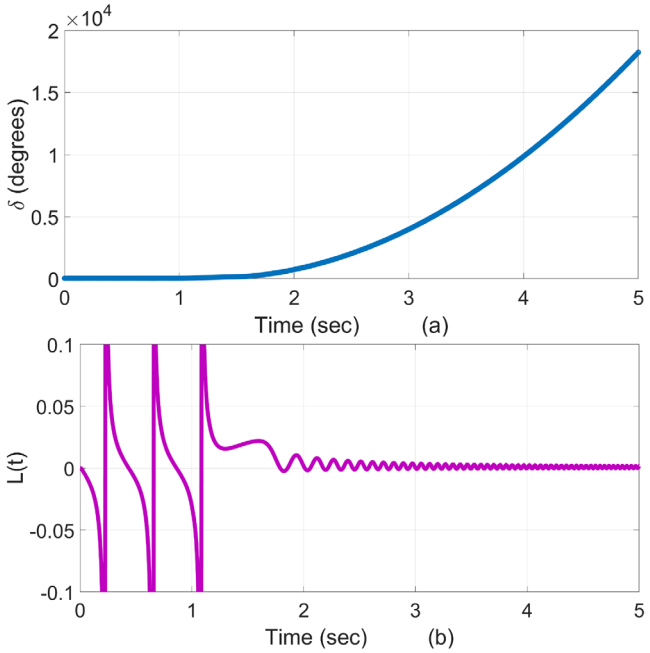


FIGURE 4 Rotor angle responses (a), solution of Riccati equation in unstable condition in SMIB (b).

6.1.2 | Unstable condition

In this section, the three-phase fault's clearance time is 0.15 s, longer than the previous case, which puts extra strain on the power system. Again, the power system is simulated numerically by Gear's method, and the corresponding DRE is integrated during the whole simulation time using the Möbius method. The power angle and solution of DRE are shown in Figure 4. Figure 4a shows that the power system is unstable since δ has an undamped exponential movement after the fault ($t = 1$ s). The solution of DRE ($L(t)$) also reflects the power system's unstable behaviour by not following the pattern of a tangent. Therefore, both sub-figures demonstrate that the system is unstable after a three-phase fault with a long clearance time of 0.15 s.

Figure 4a,b is zoomed and shown in Figure 5 to be explained more. According to Figure 5a, δ has a constant value $\delta = 47°$ during $0 \leq t < 1$ in the steady-state condition. However, it increases after the fault and passes the theoretical unstable point (conventional method) $\delta = 180°$ at $t = 1.557$ s. Therefore, the instability is detected at $t = 1.557$ s, and the simulation can be stopped in the large-signal stability analysis. Evaluating the solution of DRE ($L(t)$) shows that it is possible to detect the instability faster than the conventional method since the excursion of $L(t)$ does not follow the tangent pattern after $t = 1.35$ s. Earlier detection of instability enables the proposed method to be used to predict instability in the large-signal stability analysis. Predicting the unstable excursion before δ exceeds $\delta = 180°$ makes the large-signal stability analysis more efficient.

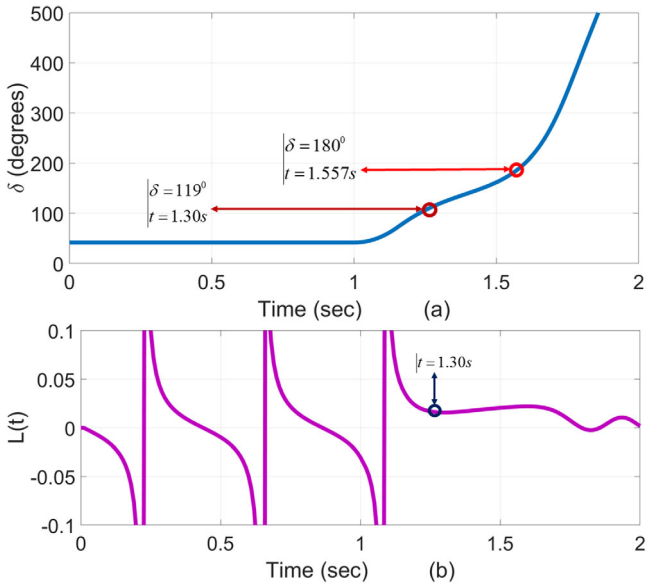


FIGURE 5 Zoomed figure of rotor angle responses and solution of Riccati equation in unstable condition in SMIB.

6.2 | Instability prediction in SMIB with AVR and PSS (sixth-order system)

The SMIB shown in Figure 2 is equipped with an automatic voltage regulator (AVR) and a power system stabilizer (PSS) to increase oscillation damping. Details of this system is presented in chapter 13 of [5]. Detailed representation of the AVR and PSS are shown in Figure 6.

In addition to the swing equation of SMIB presented in Equation (28), four complementary equations are added to the system of DAE to simulate the power system of this section. The four differential equations are:

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_{fad} = \frac{\omega_0 R_{\mathrm{fd}}}{L_{\mathrm{ads}}} E_{\mathrm{fd}} - \omega_0 R_{\mathrm{fd}} i_{\mathrm{fd}} \tag{32}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} v_1 = \frac{1}{T_{\mathrm{r}}}(E_t - v_1) \tag{33}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} v_2 = \underbrace{\frac{K_{\mathrm{stab}}}{2H_{\mathrm{int}}}(T_{\mathrm{m}} - T_{\mathrm{e}} - K_D \Delta\omega_{\mathrm{r}})}_{\mathrm{SE}} - \frac{1}{T_{\mathrm{w}}} v_2 \tag{34}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} v_{\mathrm{s}} = \frac{1}{T_2}\left(v_2 - v_{\mathrm{s}} + T_1(SE) - \frac{1}{T_{\mathrm{w}}} v_2\right) \tag{35}$$

where the excitation system is modelled by Equation (31), Equation (32) represents the AVR, and Equations (33) and (34) are related to the PSS. $R_{\mathrm{fd}} = 6 \times 10^{-4}$ and $L_{\mathrm{ads}} = 1.3726$ are the resistance and inductance of the field circuit, E_{fd} is the exciter output voltage, i_{fd} is the field current, $K_A = 200$ is the gain of the exciter, $T_{\mathrm{r}} = 0.015$ is the coefficient of the controller of the AVR, $E_{\mathrm{f_{min}}} = -6.4$ and $E_{\mathrm{f_{max}}} = +7$ are the thresholds of the limiter in the output of the PSS. $T_{\mathrm{w}} = 1.4$ is the coefficient of the washout filter, $K_{\mathrm{stab}} = 9.5$ is the gain of the PSS, $T_1 = 0.1540$

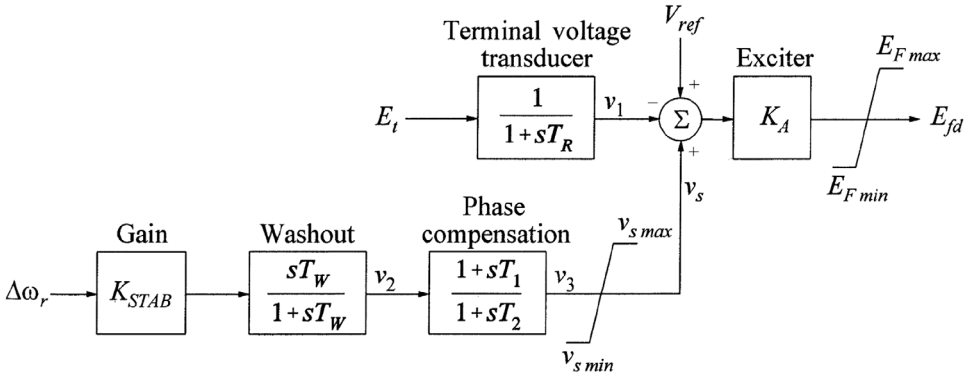


FIGURE 6 AVR-PSS.

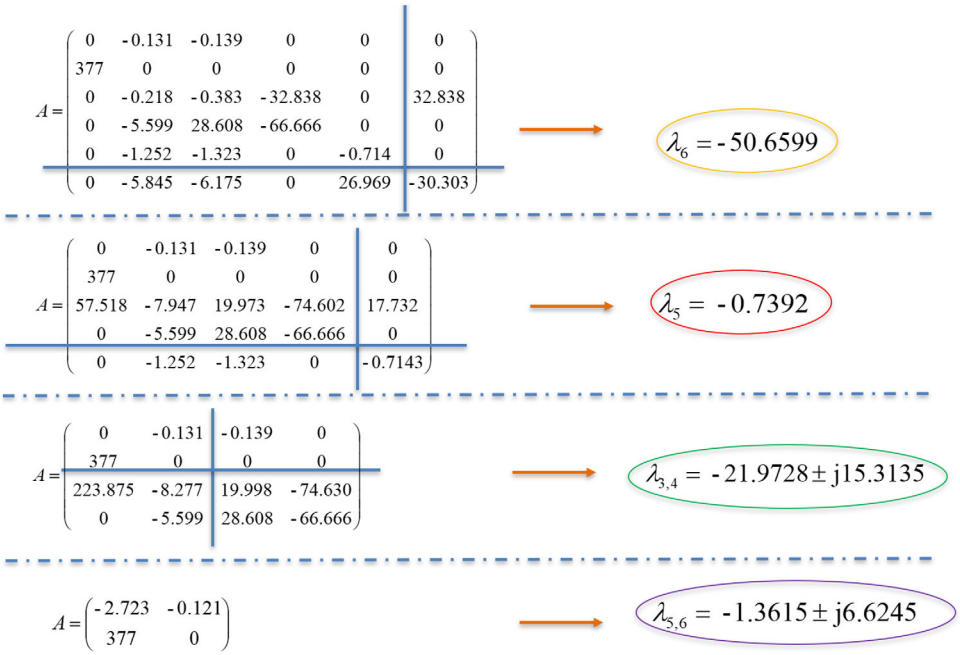


FIGURE 7 Decomposition of an A-matrix.

and $T_2 = 0.033$ are the coefficients of the controller of the PSS, $v_{s_{min}} = -0.1$ and $v_{s_{max}} = +0.1$ are the thresholds of the limiter in the output of the PSS.

The Gear's method is applied to the sixth-order system to do the power system's stability analysis. The A-matrix of the system is calculated in each integration step by taking partial differentiation of the equations with respect to the variables. Since the system is bigger than the second-order system, the procedure presented in Section 5 is used to gradually decrease the A-matrix dimension and evaluate the DRE's solution with smaller matrices individually. The A-matrix of the power system in the steady-state condition is gradually decomposed here to show the proposed procedure's performance, and the whole procedure is shown in Figure 7.

Figure 7 shows that the sixth-order A-matrix ($A_{6\times6}$) is partitioned into $A_{5\times5}$ and $A_{1\times1}$ in the first iteration. The first eigenvalue $\lambda_6 = -50.6599$ is extracted by applying V-DRE to the partitioned matrices. The smaller fifth-order A-matrix ($A_{5\times5}$) is also obtained in the first iteration, which consists of five remaining eigenvalues of the main system. The smaller matrix $A_{5\times5}$ is used as an input to the next iteration, and the same procedure (partition and make the V-DRE) is applied to extract the fifth eigenvalue ($\lambda_5 = -0.7392$) and fourth-order A-matrix. In the third iteration, A-matrix is partitioned into $A_{2\times2}$ and $A_{2\times2}$ since the eigenvalue is complex in this iteration. A second-order system is extracted by solving V-DRE whose eigenvalues are $\lambda_{3,4} = -21.9726 \pm 15.3135i$. Additionally, the remaining matrix is also 2×2 which is used to the final iteration to extract the remaining eigenvalues $\lambda_{5,6} = -1.3615 \pm 6.6245i$. Therefore, the proposed stability analysis breaks down a large matrix into the corresponding first and second-order matrices by the gradually triangularization procedure. The same procedure is used in the power system's analysis to apply the proposed index for instability prediction. The Mobius transformation is

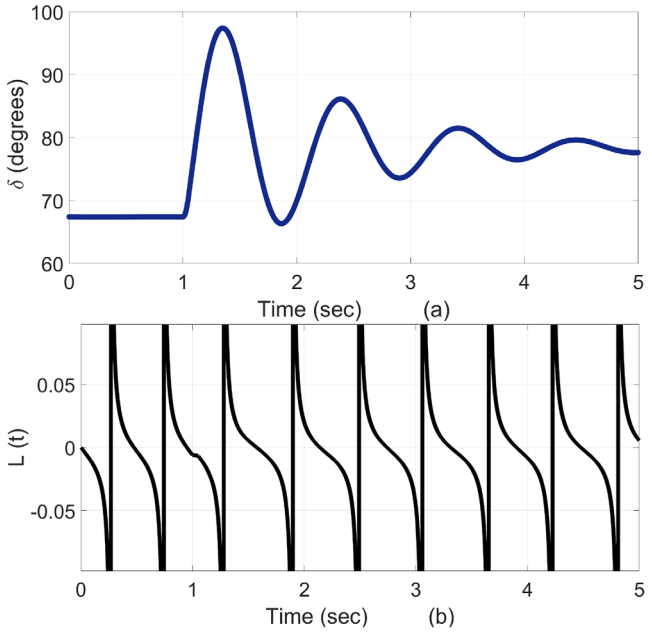


FIGURE 8 Rotor angle responses (a), and solution of DRE (b), in stable condition in SMIB+AVR+PSS.

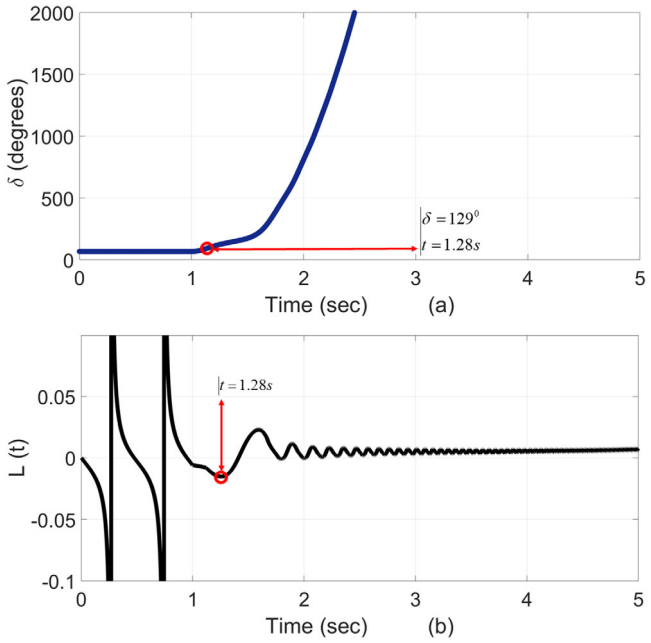


FIGURE 9 Rotor angle responses (a), and solution of DRE (b), in unstable condition in SMIB+AVR+PSS.

applied to DRE of the extracted second-order or first-order matrices to monitor the pattern of the solution of the corresponding DRE. In the following sub-sections, two cases (stable and unstable) are examined to evaluate the stability of the SMIB with AVR and PSS.

6.2.1 | Stable condition

Consider the power system (SMIB) shown in Figure 2 equipped with the AVR and the PSS shown in Figure 6. A three-phase fault occurs close to Bus2 at $t = 1$ s and is cleared by removing the line with impedance $j0.93$ with clearance time $t = 1.05$ s. The system is simulated numerically by applying the Gear's method to the sixth-order power system, and the excursion of the generator's power angle is shown in Figure 8a. The V-DRE is solved to decompose the A-matrix, and different dynamic modes of the power system are decoupled individually during steady-state and dynamic conditions. The Mobius transformation is applied to DRE of the decomposed matrix ($A_{2\times2}$) related to the mechanical part of the generator, and the solution of DRE with zero initial value ($L_0 = 0$) is shown in Figure 8b. Figure 8b shows that the system is stable since the DRE's solution follows the tangent pattern.

6.2.2 | Unstable condition

The clearing time of three-phase fault in the sixth-order power system is increased to 0.12 s in this section, and the system is simulated by the Gear's method. The power angle (δ) of the generator calculated by the Gear's method is shown in Figure 9a. According to Figure 9a, the instability is detected when δ exceeds 180° at $t = 1.57$ s. However, this paper's proposed index predicts the instability at $t = 1.3$ s by monitoring the pattern of the excursion of $L(t)$. According to the Riccati equation's solution, $L(t)$ no longer follows the tangent pattern after $t = 1.28$ s, which shows that the power system with clearance time of 0.12 s is moving toward instability.

6.3 | Instability prediction in SIMB with HVDC link

The numerical stiffness of a power system can be increased when AC and DC technologies are integrated. In the SMIB power system, the transmission line with impedance $0.5j$ is replaced with an HVDC line, and the AC–DC system (Figure 10) is simulated numerically. The HVDC consists of electrical (AC and DC) and control parts, in which the dynamic models presented in [24] are used. The electrical and the control parts of the HVDC are shown in Figure 11. Rectifier-side (Conv1) of HVDC controls the DC voltage (v_{dc1}) and reactive power (Q_1), and inverter-side (Conv2) controls the active (P_2) and the reactive power (Q_2) at the coupling buses (Bus2 and Bus3). The reference values of the HVDC are: $v_{dc1} = 1$, $Q_{ref1} = -0.186$, $P_{ref2} = 0.5853$ and $Q_{ref2} = -0.0255$ (all in per-unit), which are selected based on the steady-state condition of the SMIB without HVDC. The synchronous generator is modelled by two differential equations (the swing equations presented in Equation (28)). The AC transmission lines and the loads are also considered by the phasor models in the admittance matrix of the model. The HVDC's integration in the power system is modelled by the current injection

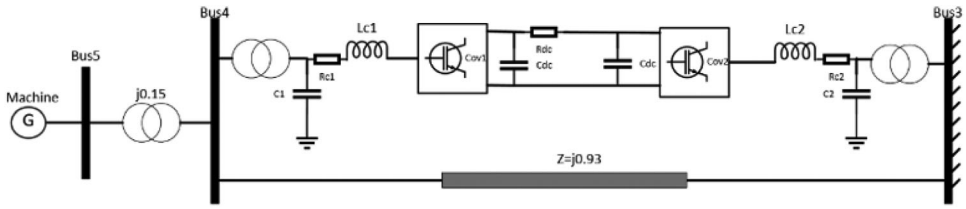


FIGURE 10 SMIB with HVDC.

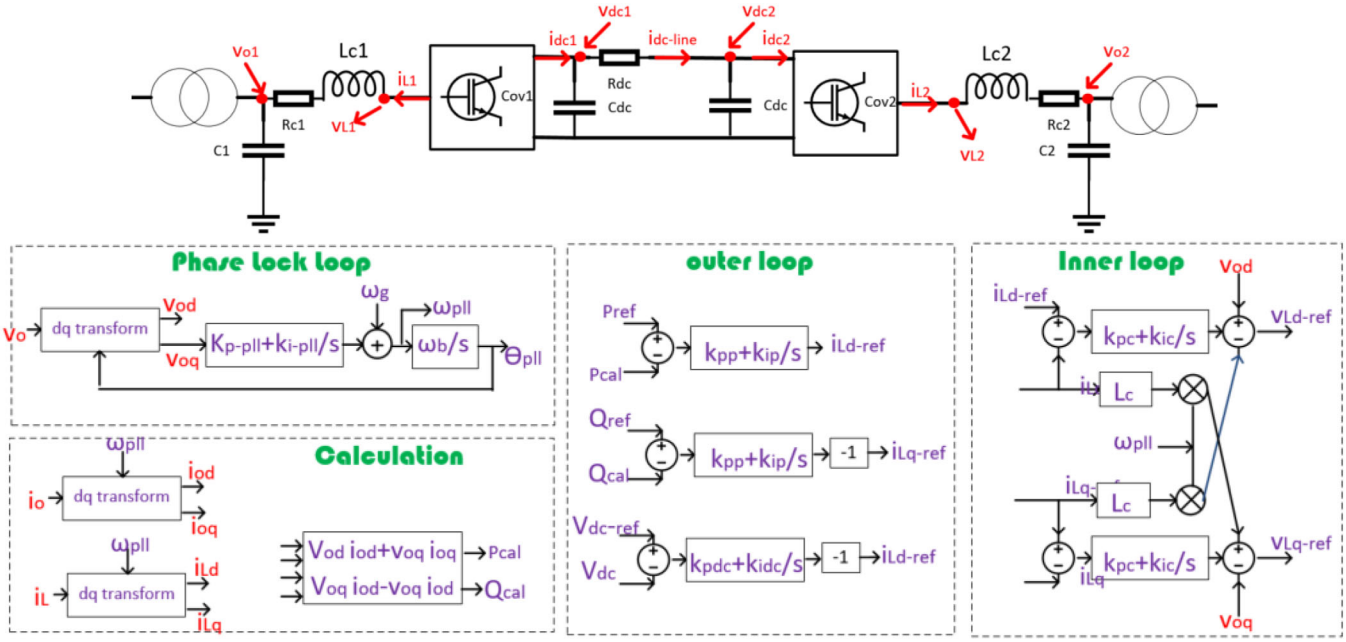


FIGURE 11 Electrical and control parts of a HVDC.

method [23, 25]. Two time-varying current sources at Bus2 and Bus3 are considered to model the HVDC link in the stability analysis. Therefore, sixteen differential equations model the whole system presented in Figure 10. Fourteen equations are related to the HVDC and two equations to the swing equations of the generator. The Gear's method is applied to the system (SMIB+HVDC), and the power system is examined in two different conditions (stable and unstable).

6.3.1 | Stable condition

The same strategy for the fault occurrence is considered in this section. The power system is under stress with a three-phase fault with a clearance time of $t = 1.05$ s. Sixteen differential equations, which model the whole power system, are considered as inputs to Gear's method, and they are integrated numerically. The result of the numerical integration for one of the state variables (δ) is shown in Figure 12a. Additionally, Gear's method extracts the A-matrix of the system during the numerical integration with dimension 16×16. The A-matrix is partitioned into $A_{2\times2}$ and $A_{14\times14}$, and the decomposed matrices are obtained by solving the vector-DRE. The decoupled $A_{2\times2}$ reflects the dynamic performance of the generator in interaction with HVDC. The Möbius method is applied to the decoupled $A_{2\times2}$, and the solution is shown in Figure 12b. According to Figure 12b, the solution starts the movement from the initial value of $L = 0$ and follows the tangent pattern during the steady-state condition ($0 \leq t < 1$). The system is still following the pattern even during oscillations ($t \geq 1$), which shows that the system is stable. Although the timing of the different tangent shapes is not the same, they still follow a general tangent pattern and so the system is table.

6.3.2 | Unstable condition

The clearance time of the fault is increased in this section to 0.1s, and the system is simulated. The excursion of δ in this condition is shown in Figure 12c. Moreover, the DRE solution corresponding to the mechanical part of the generator is also calculated by the Möbius method and is shown in Figure 12d. The simulation results show that the DRE's solution does not follow the tangent pattern when the system is unstable.

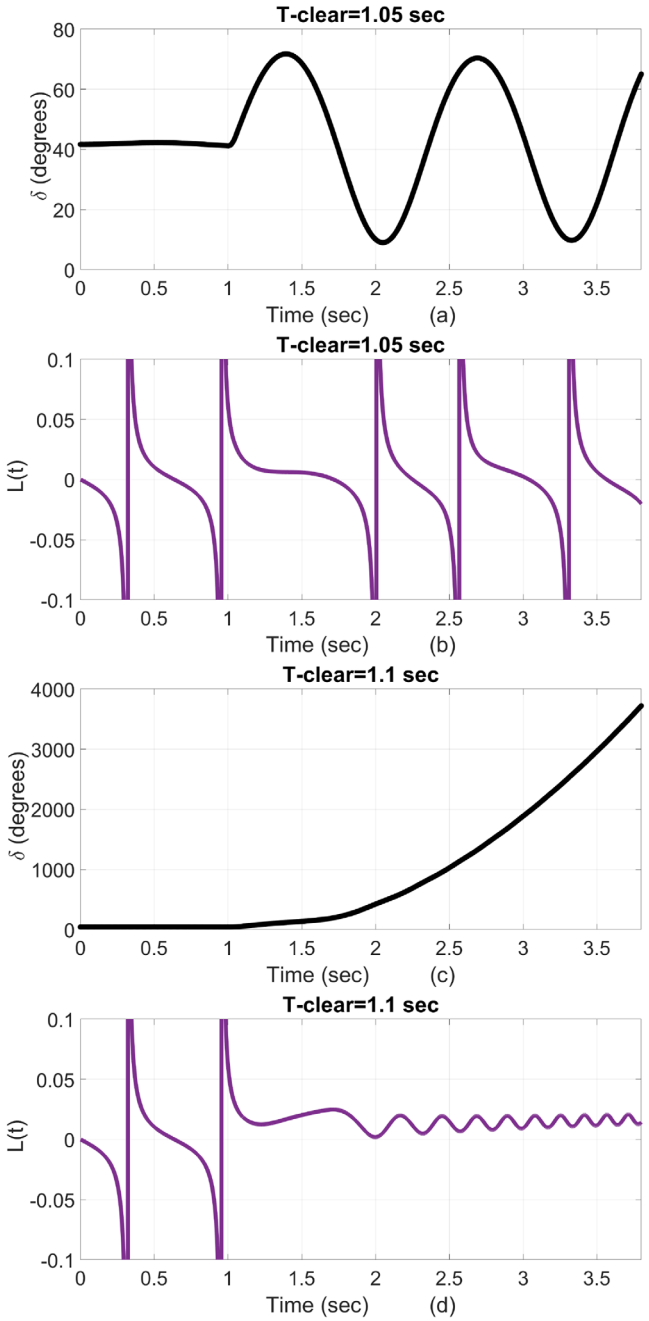


FIGURE 12 Power angle (a) and (c), and solution of Riccati equation (b) and (d), with two different clearance times (SMIB with HVDC).

Therefore, the proposed index can be used in the prediction of instability in large-signal stability analysis.

7 | DISCUSSION ON PRACTICAL APPLICABILITY

In this section, we address some practical concerns and outline potential directions for possible future research based on the methodology presented in this paper.

- Pattern recognition in DRE Solutions: Identifying whether a solution exhibits a tangent pattern is crucial for the practical application of our proposed method. To determine if a system's differential Riccati equation (DRE) solution exhibits a tangent mode, we analyse the trajectory of the solution's components, specifically looking for periodic behaviour that aligns with the characteristics of the tangent function. To enhance the accuracy and efficiency of this pattern identification, exploring advanced pattern recognition techniques, including neural networks, could be a possible research direction. This approach will help automate and refine the detection process.
- Computational demands in real-time applications: One of the primary challenges in online applications is the significant computational demand required to continuously solve the DRE and analyse its solutions in real-time. This is particularly challenging for complex power systems with high-dimensional state spaces. Interested researchers could focus on developing more streamlined computational techniques that reduce the time complexity of solving the DRE in real time. Advances in computational methods, hardware, and efficient algorithmic implementations will be pivotal in mitigating these challenges. Additionally, the development of predictive analytics to approximate the behaviour of DRE solutions could reduce the need for full-scale computation at every timestep.
- Impact of time-delay characteristics on system stability: Particularly in power systems integrated with power electronic devices, the impact of time-delay characteristics on system stability presents a complementary topic for future research. To analyse the stability of time-delay systems, augmenting the system model to reflect time-delayed states is an option. This involves modifying the A-matrix to include past states as separate state variables, which would allow the V-DRE to capture the dynamics influenced by delays.

8 | CONCLUSION

Access to the A-matrix extracted in any integration step opens new doors in the analysis of the power system's dynamic performance subjected to large signal disturbance. Classically, in large-signal stability analysis, a power system is simulated numerically, and the unstable performance is examined by visualization of the divergence of states' excursion. In this paper, a new method is proposed to make the large-signal stability analysis more efficient by predicting instability. This is done by monitoring the solution pattern of the differential Riccati equation (DRE). It is shown that the excursion of the power angle has a specific tangent pattern in stable conditions, while unstable conditions are predicted when the excursion of the state does not follow the tangent pattern. The proposed method is applied first in a simple power system (second-order) by solving the DRE. The Möbius Transformation is proposed in this paper to solve the DRE due to singularities with complex initial values. The results show that the method can pass through the singularities accurately. Secondly, the proposed instability prediction method is applied to more complex power systems (SMIB+AVR+PSS and SMIB+HVDC) by vector DRE, and

simulation results show that the proposed method is also applicable in the larger state matrix corresponding to a complex power system. Large-signal stability analysis of a power system can be stopped when the instability is predicted by the proposed method, making the whole procedure more efficient. Moreover, the proposed method can have applications in other fields of power system analysis such as digital twins, real-time power grid analysis systems, anomaly detection, and real-time simulators.

AUTHOR CONTRIBUTIONS

J. Khodaparast: Conceptualization; formal analysis; methodology; software; writing—original draft; writing—review and editing. **O. B. Fosso**: Conceptualization; funding acquisition; methodology; supervision; validation; writing—review and editing. **M. Molinas**: Conceptualization; funding acquisition; methodology; supervision; validation; writing—review and editing. **J. A. Suul**: Conceptualization; funding acquisition; methodology; supervision; validation; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets weregenerated or analysed during the current study.

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